

1

1. Two Sum

Given an array of integers `nums`, and an integer `target`, return indices of the two numbers such that they add up to `target`.

You may assume that each input would have **exactly one solution**, and you may not use the same element twice.

You can return the answer in any order.

Example 1:
Input: `nums = [2,7,11,15], target = 9`
Output: `[0,1]`
Explanation: Because `nums[0] + nums[1] == 9`, we return `[0, 1]`.

Example 2:
Input: `nums = [3,2,4], target = 6`
Output: `[1,2]`

Example 3:
Input: `nums = [3,3], target = 6`
Output: `[0,1]`

Constraints:

- $2 \leq \text{nums.length} \leq 10^5$
- $-10^9 \leq \text{nums}[i] \leq 10^9$
- $-10^9 \leq \text{target} \leq 10^9$
- Only one valid answer exists.

Follow-up: Can you come up with an algorithm that is less than $O(n^2)$ time complexity?

Seen this question in a real interview before? 1/5

Accepted 20,835,238/215,151 Acceptance Rate: 57.1%

Topics: Array, Hash Table

Companies: Google, Facebook, Microsoft, Amazon, Apple, LinkedIn, Oracle, Uber, Lyft, Airbnb, Twitter, Instagram, Pinterest, Spotify, Netflix, YouTube, Facebook, Microsoft, Amazon, Apple, LinkedIn, Oracle, Uber, Lyft, Airbnb, Twitter, Instagram, Pinterest, Spotify, Netflix, YouTube

```

class Solution {
public:
    vector<int> twoSum(vector<int>& nums, int target) {
        unordered_map<int, int> numMap;

        for (int i = 0; i < nums.size(); i++) {
            int complement = target - nums[i];
            if (numMap.find(complement) != numMap.end()) {
                return {numMap[complement], i};
            }
            numMap[nums[i]] = i;
        }

        return {};
    }
};

```

Testcase 1: Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input:

`nums = [2,7,11,15]`

`target = 9`

Output:

`[0,1]`

Expected:

`[0,1]`

Contribute a testcase

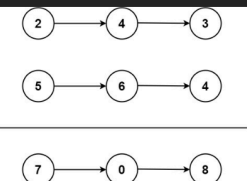
2

2. Add Two Numbers

You are given two **non-empty** linked lists representing two non-negative integers. The digits are stored in **reverse order**, and each of their nodes contains a single digit. Add the two numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

Example 1:



Input: `l1 = [2,4,3], l2 = [5,6,4]`
Output: `[7,0,8]`
Explanation: `342 + 465 = 807`.

Example 2:
Input: `l1 = [0], l2 = [0]`
Output: `[0]`

Example 3:
Input: `l1 = [9,9,9,9,9,9], l2 = [9,9,9,9]`
Output: `[0,0,0,0,0,0,1]`

Constraints:

- The number of nodes in each linked list is in the range `[1, 100]`.
- $0 \leq \text{Node.val} \leq 9$
- It is guaranteed that the list represents a number that does not have leading zeros.

Accepted 36,481/37,130 Acceptance Rate: 98.3%

Topics: Linked List, Math

Companies: Google, Facebook, Microsoft, Amazon, Apple, LinkedIn, Oracle, Uber, Lyft, Airbnb, Twitter, Instagram, Pinterest, Spotify, Netflix, YouTube

```

class Solution {
public:
    ListNode* addTwoNumbers(ListNode* l1, ListNode* l2) {
        ListNode* dummyhead = new ListNode(0);
        ListNode* current = dummyhead;
        int carry = 0;

        while (l1 != nullptr || l2 != nullptr || carry != 0) {
            int sum = carry;
            if (l1 != nullptr) {
                sum += l1->val;
                l1 = l1->next;
            }
            if (l2 != nullptr) {
                sum += l2->val;
                l2 = l2->next;
            }
            carry = sum / 10;
            current->next = new ListNode(sum % 10);
            current = current->next;
        }

        ListNode* result = dummyhead->next;
        delete dummyhead;
        return result;
    }
};

```

Testcase 1: Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input:

`l1 = [2,4,3]`

`l2 = [5,6,4]`

Output:

`[7,0,8]`

Expected:

`[7,0,8]`

3

leetcode.com/problems/longest-substring-without-repeating-characters/

3. Longest Substring Without Repeating Characters

Given a string *s*, find the length of the **longest substring** without duplicate characters.

Example 1:
Input: *s* = "abcabcbb"
Output: 3
Explanation: The answer is "abc", with the length of 3. Note that "bca" and "cab" are also correct answers.

Example 2:
Input: *s* = "bbbbb"
Output: 1
Explanation: The answer is "b", with the length of 1.

Example 3:
Input: *s* = "pwwkew"
Output: 3
Explanation: The answer is "wke", with the length of 3. Notice that the answer must be a substring, "pke" is a subsequence and not a substring.

Constraints:

- 0 ≤ *s*.length ≤ 5 × 10⁴
- s* consists of English letters, digits, symbols and spaces.

Seen this question in a real interview before? 1/5

Accepted 9,052,023/23.5M Acceptance Rate 38.5%

Topics: Array, Hash Table, String, Sliding Window

Companies: Amazon, Microsoft, Facebook, Google, Apple, LinkedIn, Uber, etc.

Hint 1: Generate all possible substrings & check for each substring if it's valid and keep updating maxLen accordingly.

Similar Questions: Longest Substring with At Most Two Distinct Characters, Longest Substring with At Most K Distinct Characters, Longest Substring Without Repeating Characters II

Discussion 824

44.6K 824 1 1 1

831 Online

```
class Solution {
public:
    int lengthOfLongestSubstring(string s) {
        unordered_map<char, int> charmap;
        int maxlength = 0;
        int start = 0;

        for (int end = 0; end < s.length(); end++) {
            if (charmap.find(s[end]) != charmap.end()) {
                start = max(start, charmap[s[end]] + 1);
            }
            charmap[s[end]] = end;
            maxlength = max(maxlength, end - start + 1);
        }

        return maxlength;
    }
};
```

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input: s = "abcabcbb"

Output: 3

Expected: 3

Contribute a testcase

4

leetcode.com/problems/median-of-two-sorted-arrays/

4. Median of Two Sorted Arrays

Given two sorted arrays *nums1* and *nums2* of size *m* and *n* respectively, return the **median** of the two sorted arrays. The overall run time complexity should be $O(\log(m+n))$.

Example 1:
Input: *nums1* = [1,3], *nums2* = [2]
Output: 2.00000
Explanation: merged array = [1,2,3] and median is 2.

Example 2:
Input: *nums1* = [1,2], *nums2* = [3,4]
Output: 2.50000
Explanation: merged array = [1,2,3,4] and median is (2 + 3) / 2 = 2.5.

Constraints:

- nums1*.length ≤ 100
- nums2*.length ≤ 100
- 0 ≤ *nums1*[i] ≤ 10000
- 0 ≤ *nums2*[i] ≤ 10000
- 1 ≤ *m* + *n* ≤ 2000
- $\text{ceil}(\text{len}(\text{nums1}) + \text{len}(\text{nums2}) / 2)$ is odd

Seen this question in a real interview before? 1/5

Accepted 4,082,614/4.1M Acceptance Rate 46.0%

Topics: Array, Binary Search

Companies: Amazon, Microsoft, Facebook, Google, Apple, LinkedIn, Uber, etc.

Hint 1: If *nums1* and *nums2* are merged, then the median will be the middle element of the merged array.

Similar Questions: Find Median from Data Stream, Find Median of Two Sorted Arrays II, Find Median of Two Sorted Arrays III

Discussion 384

23.9K 384 1 1 1

482 Online

```
class Solution {
public:
    double findMedianSortedArrays(vector<int>& nums1, vector<int>& nums2) {
        int m = nums1.size();
        int n = nums2.size();
        int total = (m + n + 1) / 2;

        int left = 0, right = m;
        while (left < right) {
            int i = (left + right) / 2;
            int j = total - i - 1;

            int nums1_i = (i < m ? nums1[i] : INT_MIN);
            int nums1_j = (j < m ? nums1[j] : INT_MIN);
            int nums2_j = (j < n ? nums2[j] : INT_MIN);
            int nums2_i = (i < n ? nums2[i] : INT_MIN);

            if (nums1_i <= nums2_j) {
                left = i + 1;
            } else {
                right = i;
            }
        }

        int i = left;
        int j = total - i - 1;

        if ((m + n) % 2 == 1) {
            return nums1[i];
        } else {
            return (nums1[i] + nums2[j]) / 2.0;
        }
    }
};
```

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input: nums1 = [1,3], nums2 = [2]

Output: 2.00000

Expected: 2.00000

Contribute a testcase

5

5. Longest Palindromic Substring

Medium Topics Companies Hint

Given a string s , return the longest palindromic substring in s .

Example 1:
Input: $s = \text{"babad"}$
Output: "bab"
Explanation: "aba" is also a valid answer.

Example 2:
Input: $s = \text{"cbbd"}$
Output: "bb"

Constraints:

- $1 \leq s.length \leq 1000$
- s consist of only digits and English letters.

See this question in a real interview before? 1/5

Accepted 4,560,078/12.2M Acceptance Rate 37.4%

Topics Companies

Hint 1
How can we reuse a previously computed palindrome to compute a larger palindrome?

Hint 2
If "aba" is a palindrome, is "ababa" a palindrome? Similarly is "abab" a palindrome?

Hint 3
Complexity based hint:
If we use brute force and check whether for every start and end position a substring is a palindrome we have $O(n^2)$ start - end pairs and $O(n)$ palindromic checks. Can we reduce the time for palindromic checks to $O(1)$ by reusing some previous computation.

Similar Questions

32.4K 581 538 Online

```
class Solution {
public:
    string longestPalindrome(string s) {
        if (s.empty()) return "";

        int start = 0, maxlen = 1;

        for (int i = 0; i < s.length(); i++) {
            expandaroundcenter(s, i, start, maxlen);
            expandaroundcenter(s, i + 1, start, maxlen);
        }

        return s.substr(start, maxlen);
    }

private:
    void expandaroundcenter(const string& s, int left, int right, int& start, int& maxlen) {
        while (left >= 0 && right < s.length() && s[left] == s[right]) {
            left--;
            right++;
        }

        int len = right - left - 1;
        if (len > maxlen) {
            maxlen = len;
            start = left + 1;
        }
    }
};
```

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2

Input

$s =$

"babad"

Output

"bab"

Expected

"bab"

Contribute a testcase

6

6. Zigzag Conversion

Medium Topics Companies

The string "PAYPALISHIRING" is written in a zigzag pattern on a given number of rows like this (you may want to display this pattern in a fixed font for better legibility)

P A H N
A P L S I G
Y I R

And then read line by line: "PAHNAPLSIIGYIR"

Write the code that will take a string and make this conversion given a number of rows:

$\text{string convert(string s, int numRows);}$

Example 1:
Input: $s = \text{"PAYPALISHIRING"}$, $\text{numRows} = 3$
Output: "PAHNAPLSIIGYIR"

Example 2:
Input: $s = \text{"PAYPALISHIRING"}$, $\text{numRows} = 4$
Output: "PINALSIGYAHRPI"
Explanation:
P I N
A L S I G
Y A H R
P I

Example 3:
Input: $s = \text{"A"}$, $\text{numRows} = 1$
Output: "A"

Constraints:

- $1 \leq s.length \leq 1000$
- s consists of English letters (lower-case and upper-case), " " , and "." .
- $1 \leq \text{numRows} \leq 1000$

See this question in a real interview before? 1/5

Accepted 2,117,743/4M Acceptance Rate 53.6%

9.3K 531 500 Online

```
class Solution {
public:
    string convert(string s, int numRows) {
        if (numRows == 1 || s.length() <= numRows) {
            return s;
        }

        vector<string> rows(numRows);
        int currentrow = 0;
        bool goingdown = false;

        for (char c : s) {
            rows[currentrow] += c;

            if (currentrow == 0 || currentrow == numRows - 1) {
                goingdown = !goingdown;
            }

            currentrow += goingdown ? 1 : -1;
        }

        string result;
        for (const string& row : rows) {
            result += row;
        }

        return result;
    }
};
```

Testcase Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

$s =$

"PAYPALISHIRING"

$\text{numRows} =$

3

Output

"PAHNAPLSIIGYIR"

Expected

"PAHNAPLSIIGYIR"

Contribute a testcase

7

8